Complex Social &
Computational Systems

NEURAL NETWORKS II — LARGE LANGUAGE MODELS

A PSYCHOLOGIST'S GUIDE TO MACHINE LEARNING

ZURICH, MARCH 16-20, 2026

“generating text”

“learned from data and can be fine-tuned”

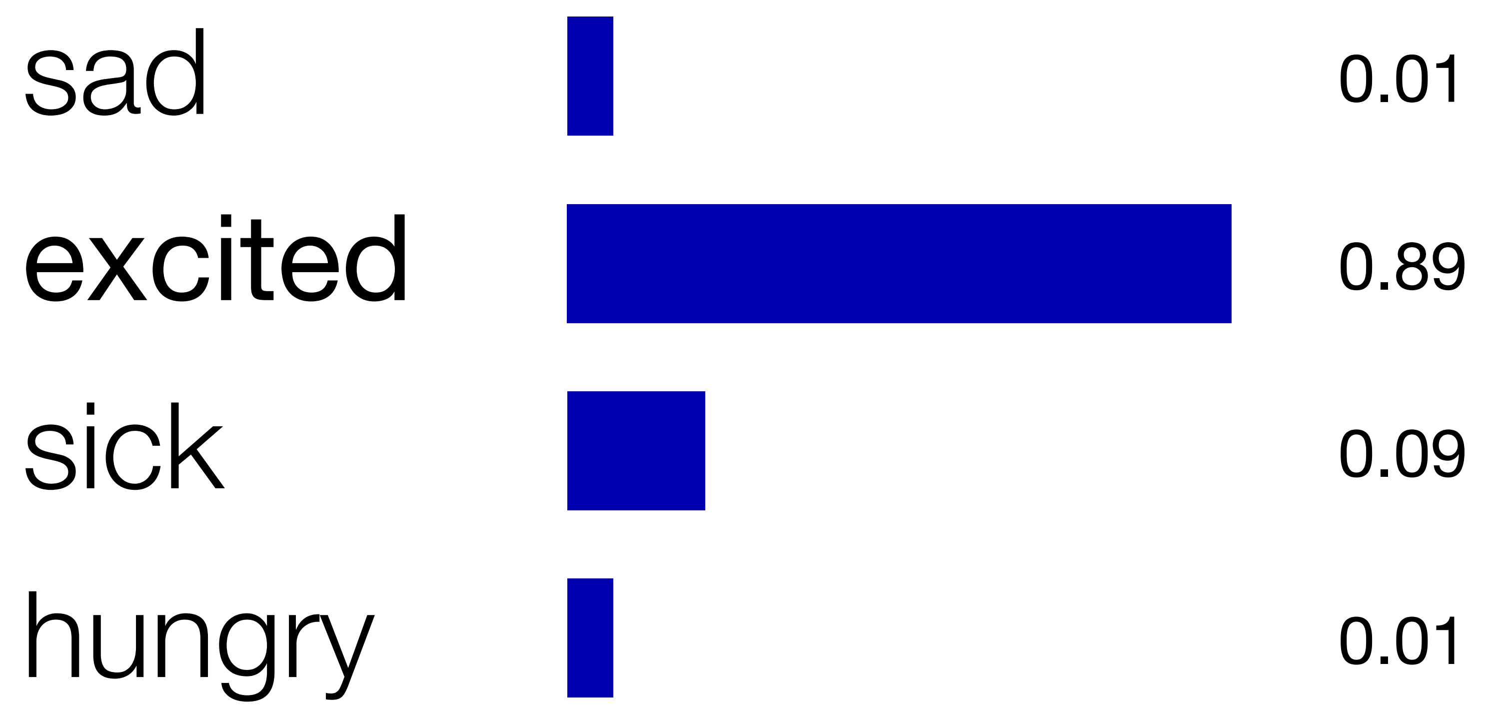
GENERATIVE PRE-TRAINED TRANSFORMERS

the name of the machine learning architecture

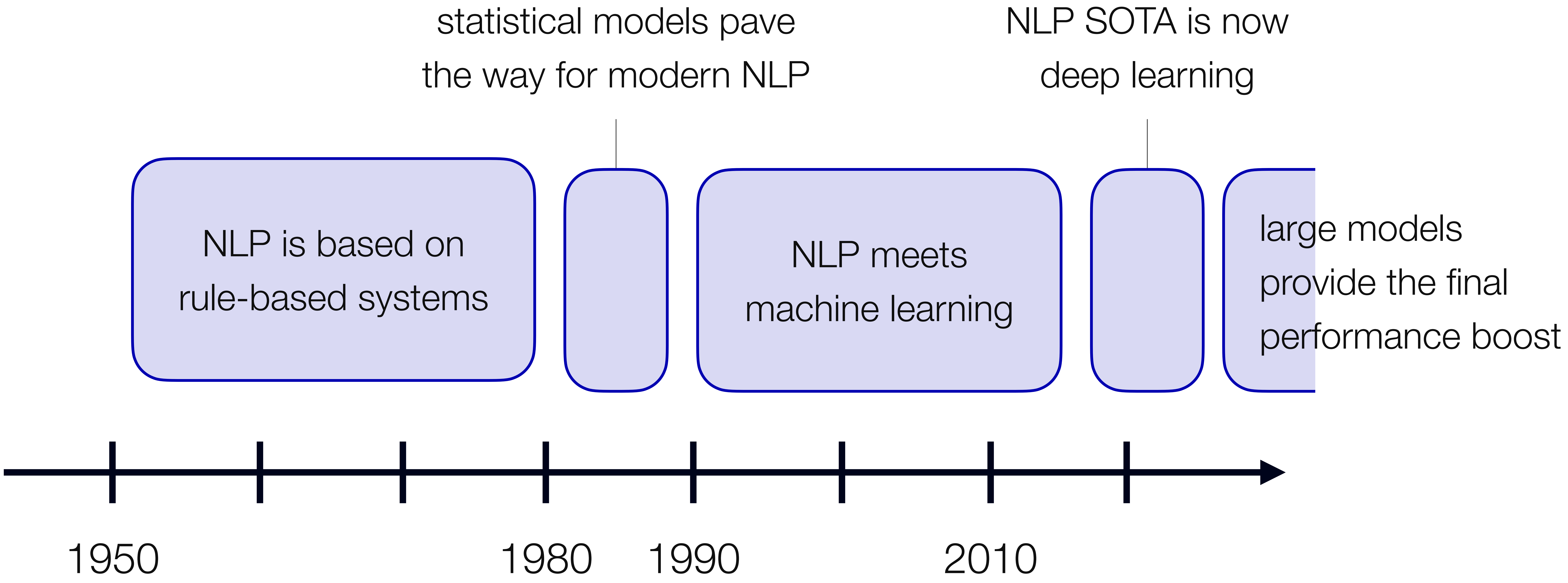
GPTS: SELF-SUPERVISED LEARNING FOR SEQUENCE PREDICTION.

Goal: Given a sequence of words, predict the next word.

"I bought tickets for my favorite band, who are in town on the weekend. I am already very …"



NATURAL LANGUAGE PROCESSING HAS COME A LONG WAY.



THE BREAK THROUGH: ATTENTION IS ALL YOU NEED.

Attention Is All You Need

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(Vaswani et al., 2017)

INSIDE GPTS: WHAT IS ATTENTION AND HOW DOES IT WORK?

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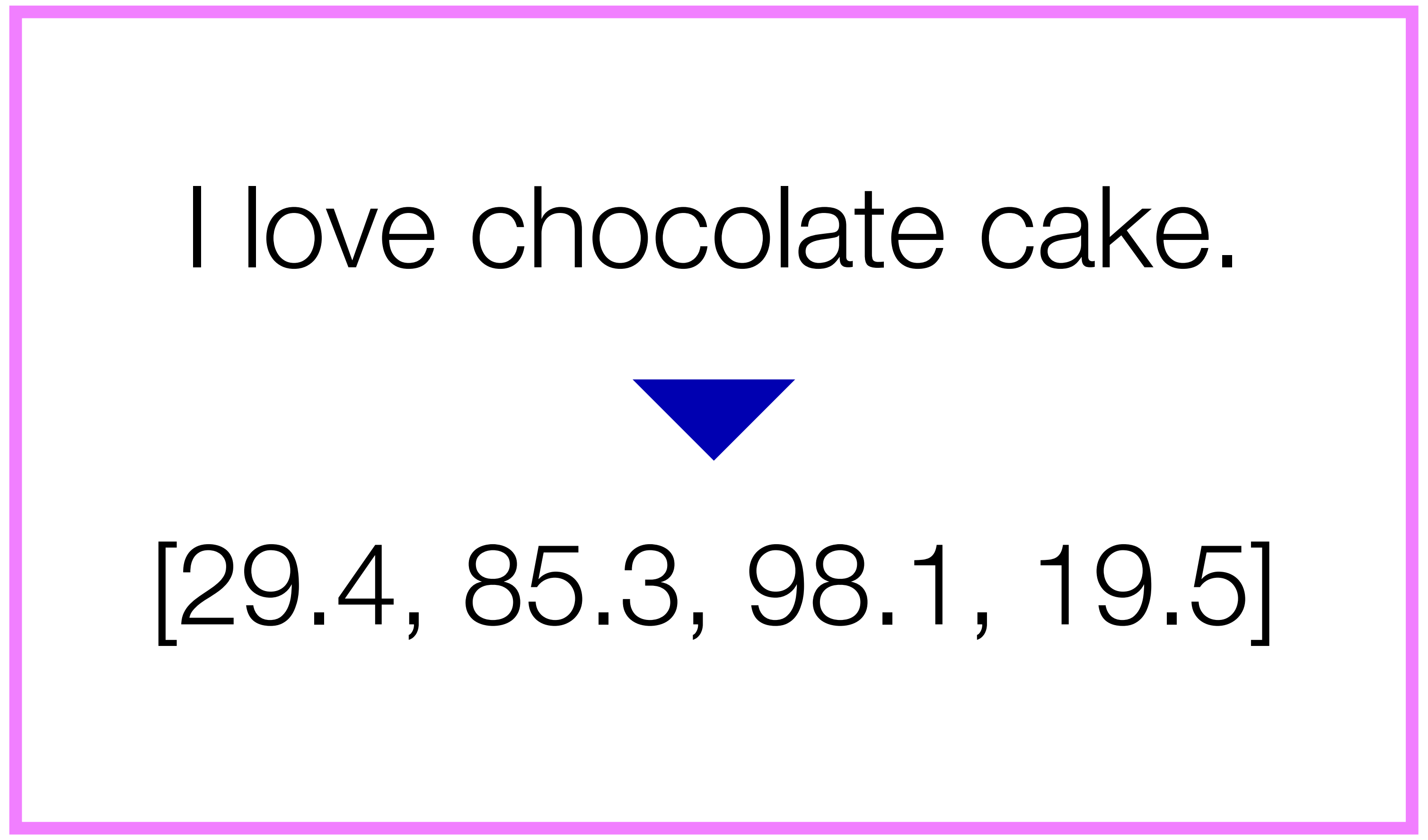
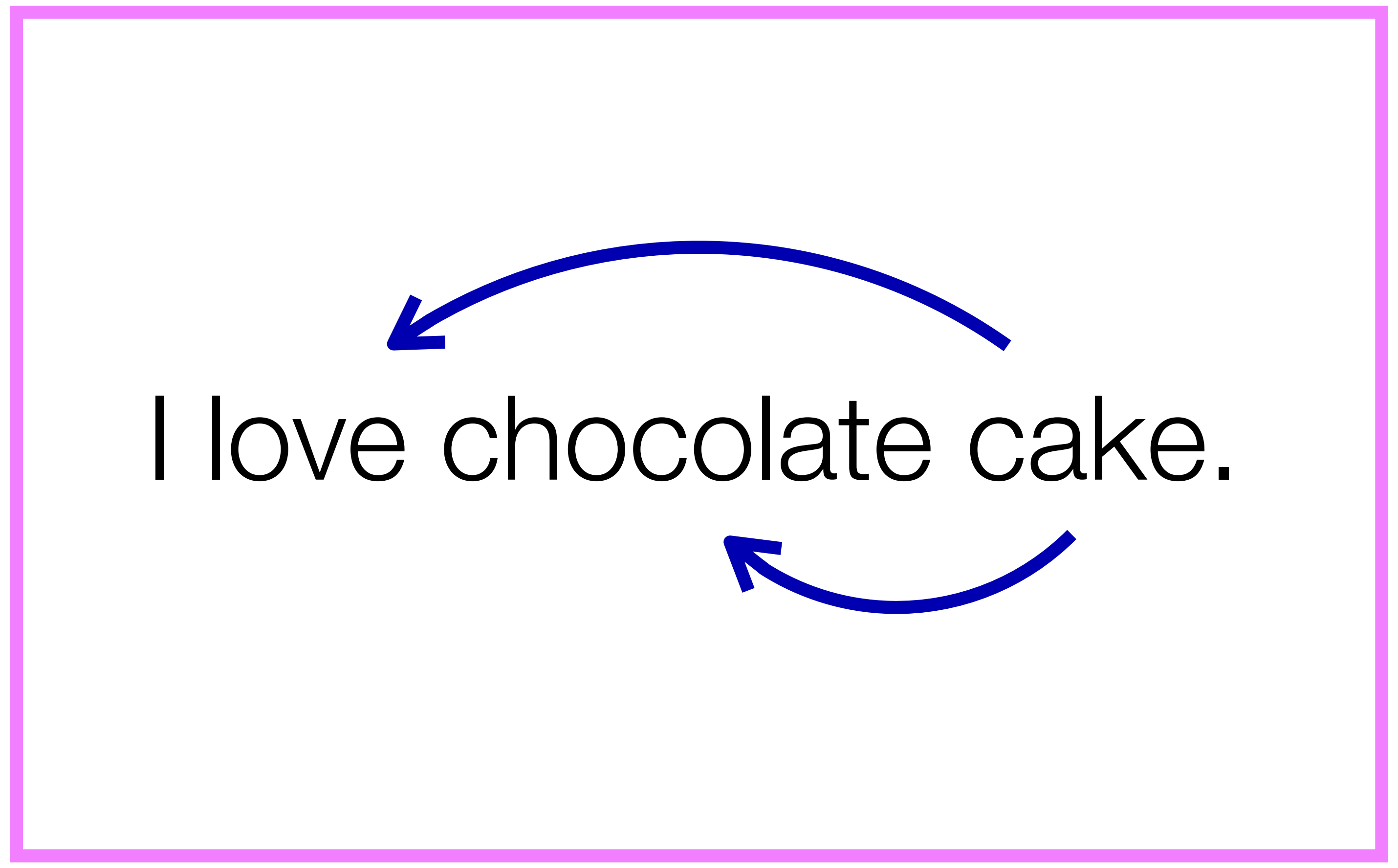
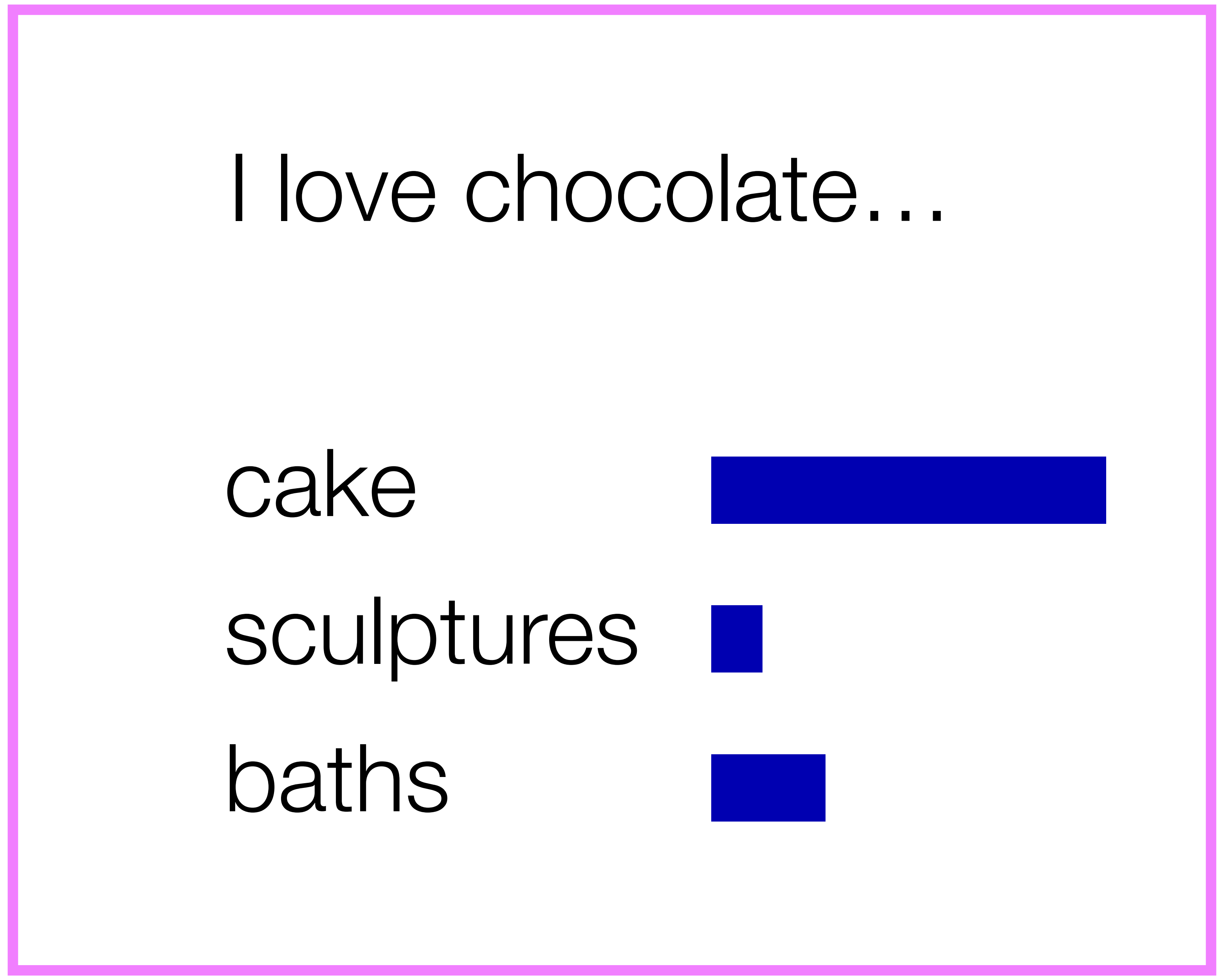


WORDS TO NUMBERS



ATTENTION



NEXT TOKEN PREDICTION

EXCURSUS: WORDS AS NUMBERS — WHERE IT STARTED.

"I love chocolate cake."

"I drink coffee with milk."

love	chocolate	cake	drink	coffee	milk
1	0	0	0	0	0
0	1	0	0	0	0
0	0	1	0	0	0
0	0	0	1	0	0
0	0	0	0	1	0
0	0	0	0	0	1

EXCURSUS: WORDS AS NUMBERS – WHERE IT STARTED.

"I love chocolate cake."

"I drink coffee with milk."

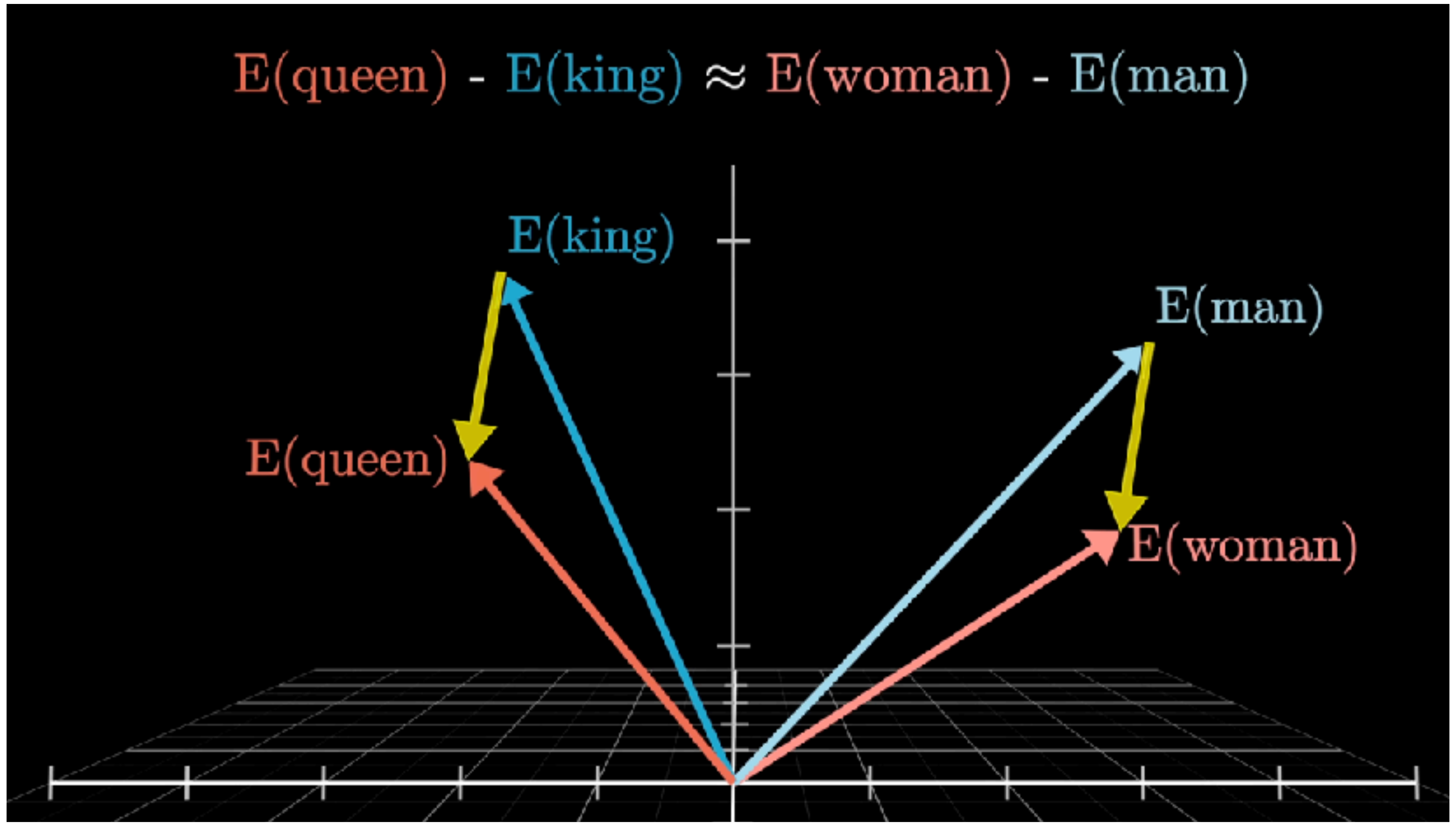
	love	chocolate	cake	drink	coffee	milk
"I love chocolate cake."	1	1	1	0	0	0
"I drink coffee with milk."	0	0	0	1	1	1

Adjust based on frequency of word in sentence and frequency of word in corpus.

WORDS AS NUMBERS TODAY: TOKENIZATION & WORD EMBEDDINGS.

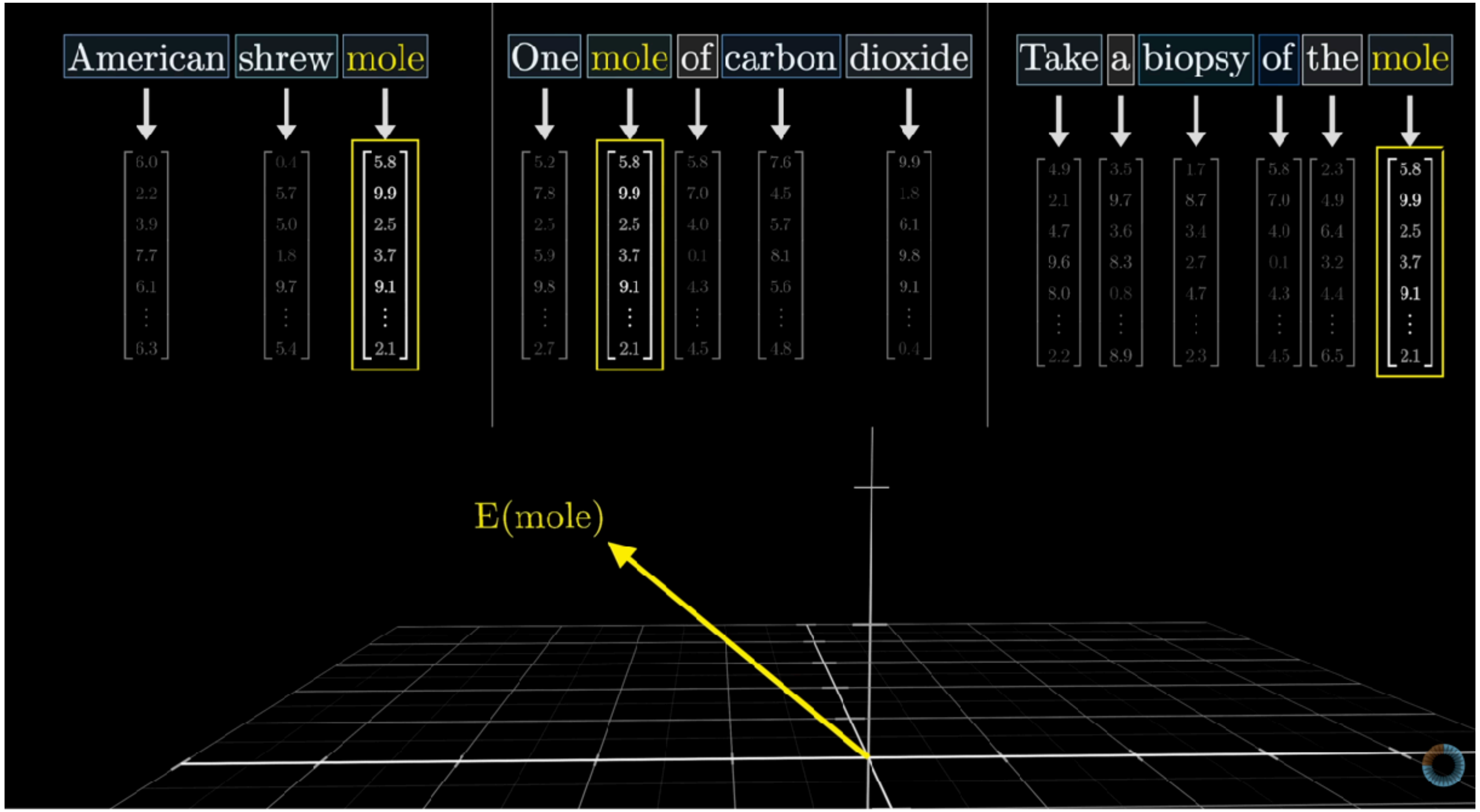
Masked language modeling: I love **[MASK]** cake.

I	love	chocolate	cake
$\begin{bmatrix} 1.3 \\ 4.5 \\ 2.9 \\ 7.6 \\ \dots \\ 8.0 \end{bmatrix}$	$\begin{bmatrix} 1.3 \\ 4.5 \\ 2.9 \\ 7.6 \\ \dots \\ 8.0 \end{bmatrix}$	$\begin{bmatrix} 1.3 \\ 4.5 \\ 2.9 \\ 7.6 \\ \dots \\ 8.0 \end{bmatrix}$	$\begin{bmatrix} 1.3 \\ 4.5 \\ 2.9 \\ 7.6 \\ \dots \\ 8.0 \end{bmatrix}$



3Blue1Brown, Transformers, the tech behind LLMs | Deep Learning Chapter 5

WORDS AS NUMBERS TODAY: TOKENIZATION & WORD EMBEDDINGS.

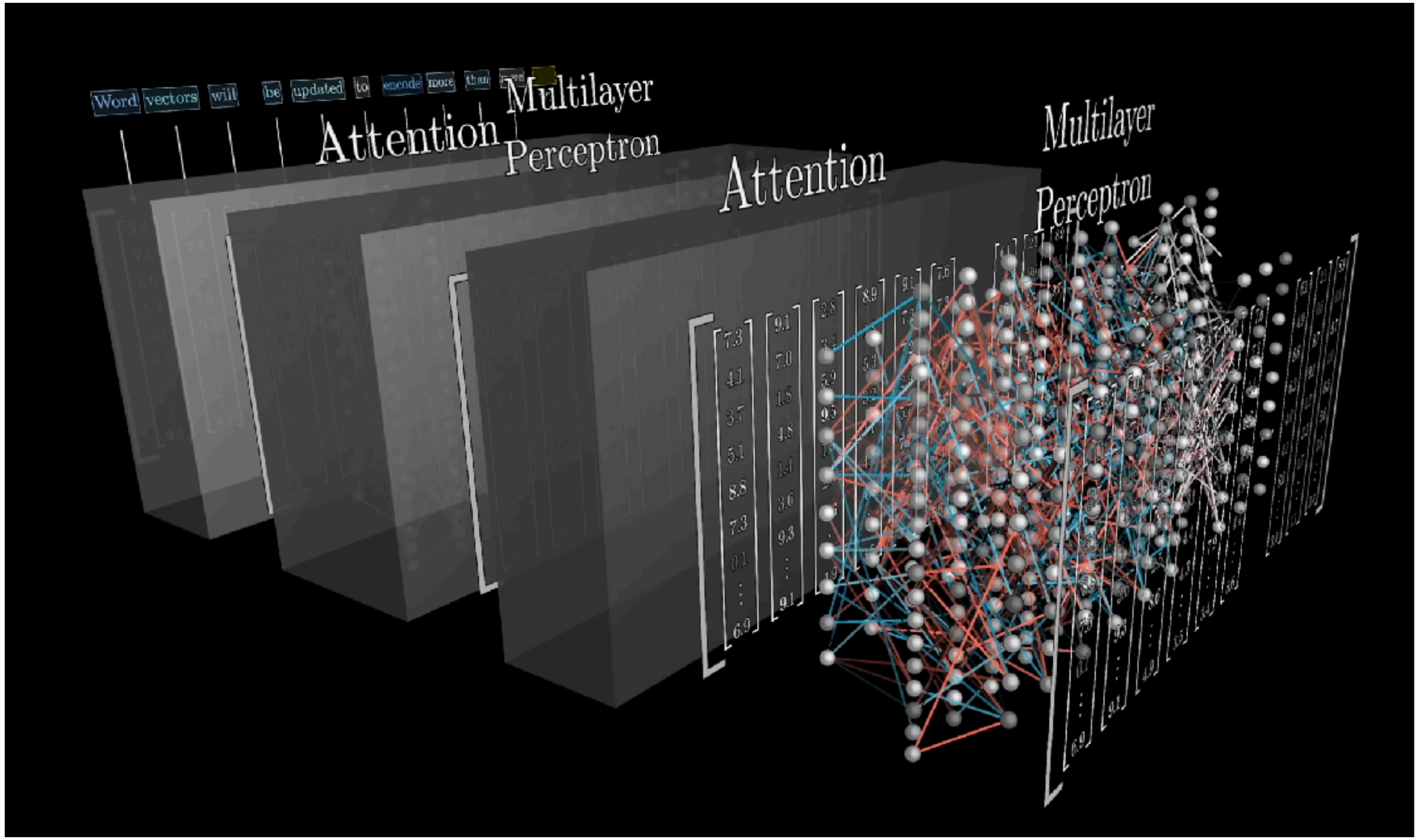


3Blue1Brown, Attention in transformers, step-by-step | Deep Learning Chapter 6

CREATING UNDERSTANDING IN LANGUAGE MODELS: ATTENTION.

	I	drink	coffee	with	milk.
I	x	●	●	●	●
drink	x	x	●	●	●
coffee	x	x	x	●	●
with	x	x	x	x	●
milk	x	x	x	x	x

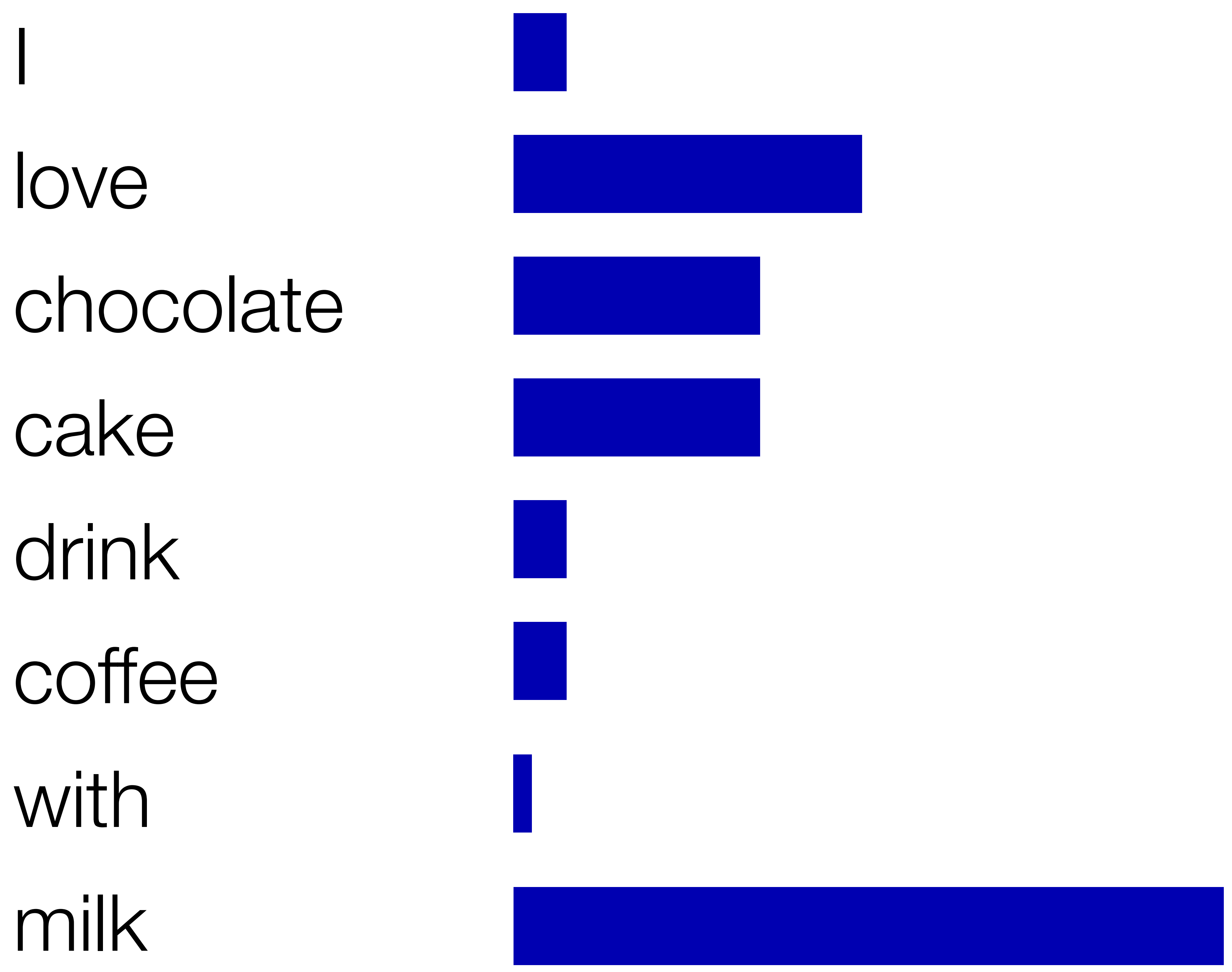
CREATING UNDERSTANDING IN LANGUAGE MODELS: STACK IT UP AGAIN.



3Blue1Brown, Transformers, the tech behind LLMs | Deep Learning Chapter 5

TEXT GENERATION: PREDICTING THE MOST LIKELY NEXT WORD.

I drink coffee with …



The output vector is as large as the vocabulary.

TODAY THERE ARE MANY DIFFERENT POWERFUL LLMS.

Proprietary versus Open Source/Open Weights models

Proprietary	Open Source/Open Weights
GPT (OpenAI)	Qwen (Alibaba)
Claude (Anthropic)	Kimi (Moonshot AI)
Gemini (Google)	Mistral (Mistral)
Grok (xAI)	Llama (Meta)

A WORD ON PROMPTING

PROMPT ENGINEERING IS THE REFINEMENT OF PROMPTS GIVEN TO LLMS TO ACHIEVE A CERTAIN TASK.

HOW TO PROMPT ENGINEERING: CHAIN OF THOUGHT.

I need you to label the following text as hate speech or not.

Here is a definition: Hateful speech includes insults, discrimination, or intimidation of individuals or groups on the Internet, on the grounds of their supposed race, ethnic origin, gender, religion, or political beliefs. It also includes speech that aims to spread fearful, negative, and harmful stereotypes, call for exclusion or segregation, incite hatred, and encourage violence against a particular group.

Please label the following text: "Gabriel the fat mastiff. Should rather go build refugee homes with Til Mettigel Schweiger. Fat son of a bitch."

Please provide a reason before making a final decision.

ChatGPT

Reasoning

The text contains **insults and abusive language** directed at a specific individual ("Gabriel"), including phrases like "fat mastiff" and "fat son of a bitch." These are **personal insults.**

However, according to the provided definition, *hate speech* must target individuals or groups **based on protected characteristics** such as race, ethnic origin, gender, religion, or political beliefs, or promote harmful stereotypes, exclusion, or violence toward such groups.

In this text:

- The insults focus on **body weight** and general abuse.
- The reference to "build refugee homes" does not clearly insult refugees as a group; it appears more like a sarcastic remark about the individual mentioned.
- No protected group (race, religion, ethnicity, etc.) is directly attacked.

Therefore, the content is **offensive/abusive**, but it does **not meet the definition of hate speech** given.

Final label: Not hate speech.

HOW TO PROMPT ENGINEERING: FEW SHOT LEARNING.

I need you to label the following text as hate speech or not.

Here is a definition: Hateful speech includes insults, discrimination, or intimidation of individuals or groups on the Internet, on the grounds of their supposed race, ethnic origin, gender, religion, or political beliefs. It also includes speech that aims to spread fearful, negative, and harmful stereotypes, call for exclusion or segregation, incite hatred, and encourage violence against a particular group.

Here are a few examples of hate speech:
- "Filthy Lying Media high traitors go to hell."
- "The Green Party is as unnecessary as athlete's foot."
- "Women aren't capable of doing politics."

Please label the following text: Gabriel the fat mastiff. Should rather go build refugee homes with Til Mettigel Schweiger. Fat son of a bitch.

Please provide a reason before making a final decision.

ChatGPT

Reasoning

The text contains **strong personal insults**, such as "fat mastiff" and "fat son of a bitch," directed at an individual named Gabriel. This language is abusive and derogatory.

However, according to the provided definition, **hate speech must target individuals or groups based on protected characteristics** (e.g., race, ethnic origin, gender, religion, or political beliefs) or promote harmful stereotypes, exclusion, or violence toward such groups.

In this text:

- The insults are **directed at a specific person**, not at a protected group.
- The mention of "build refugee homes" appears to be a **sarcastic jab**, but it does not directly insult refugees or call for discrimination against them.
- There is **no explicit attack on a protected group or characteristic** as defined in the instructions.

Therefore, while the statement is **offensive and abusive**, it **does not meet the definition of hate speech** provided.

Final label: Not hate speech.

HOW TO PROMPT ENGINEERING: INSTRUCTIONS ON OUTPUT FORMAT.

I need you to label the following text as hate speech or not.

Here is a definition: Hateful speech includes insults, discrimination, or intimidation of individuals or groups on the Internet, on the grounds of their supposed race, ethnic origin, gender, religion, or political beliefs. It also includes speech that aims to spread fearful, negative, and harmful stereotypes, call for exclusion or segregation, incite hatred, and encourage violence against a particular group.

Here are a few examples of hate speech:
- "Filthy Lying Media high traitors go to hell."
- "The Green Party is as unnecessary as athlete's foot."
- "Women aren't capable of doing politics."

Please label the following text: Gabriel the fat mastiff. Should rather go build refugee homes with Til Mettigel Schweiger. Fat son of a bitch.

Please provide a reason before making a final decision. At the end, please answer in the following format: "Label: {you label}". Use 1 for hate speech and 0 for not hate speech.

Reasoning

The text contains **insulting and abusive language** directed at a specific individual named Gabriel, including phrases such as "fat mastiff" and "fat son of a bitch." These expressions attack the person's **appearance/body weight** and use general profanity.

According to the provided definition, hate speech must involve **attacks or discrimination against individuals or groups based on protected characteristics**, specifically race, ethnic origin, gender, religion, or political beliefs, or it must promote stereotypes, exclusion, or violence toward such groups.

In this case:

- The insults are **personal attacks,** not directed at a protected group.
- The phrase about "build refugee homes" appears to be **sarcastic or mocking**, but it does not explicitly insult refugees as a group or advocate discrimination or violence against them.
- There is **no clear targeting of a protected characteristic** listed in the definition.

Therefore, the statement qualifies as **abusive language**, but **not hate speech** under the given definition.

Label: 0

PROMPT ENGINEERING — MORE AN ART THAN A SCIENCE.

Researchers have observed curious behavior between semantically identical prompts. For example, performance of LLMs can vary when

… using different **synonyms** like “excludes/lacks” or “classify/categorize”.

… using **American versus British English** (e.g., “categorize” versus “categorise”)

… using **alternative**, but semantically equivalent **phrasings**.

(Pichler et al., 2025; Mizrahi et al., 2024; after Steiner, 2026 — Lecture on “Prompt and Context Engineering”)

WHEN LLMS ARE (NOT) SAFE TO USE.

LLMS STRUGGLE WITH CERTAIN TASKS.

- Arithmetic and maths
- Spelling and counting
- Web search and very recent information
- Spatial, temporal and logical reasoning
- Disagreeing with the user

(Adapted from Steiner, 2026 — Lecture on “Prompt and Context Engineering”)

LLMS ARE GREAT CODING SPARRING PARTNERS.

Can you write Python code for me to display the results of a t-test computed with scipy in a bar chart including confidence intervals?

Python

Run

```python
import numpy as np
import matplotlib.pyplot as plt
from scipy import stats

# Example data
group1 = np.array([12, 15, 14, 10, 13])
group2 = np.array([18, 20, 17, 22, 19])

# Run t-test
t_stat, p_value = stats.ttest_ind(group1, group2)

print(f"t = {t_stat:.3f}, p = {p_value:.3f}")

# Compute means
means = [np.mean(group1), np.mean(group2)]

# Standard error of the mean
sem1 = stats.sem(group1)
sem2 = stats.sem(group2)

# Degrees of freedom
df1 = len(group1) - 1
df2 = len(group2) - 1
```

Save lines by giving means to plotting function directly

```python
# 95% confidence intervals
ci1 = stats.t.interval(0.95, df1, loc=means[0], scale=sem1)
ci2 = stats.t.interval(0.95, df2, loc=means[1], scale=sem2)

# Convert CI to error bar size
errors = [
    means[0] - ci1[0],
    means[1] - ci2[0]
]

# Bar chart
labels = ["Group 1", "Group 2"]

plt.bar(labels, means, yerr=errors, capsize=8)
plt.ylabel("Mean value")
plt.title(f"T-test result: t = {t_stat:.2f}, p = {p_value:.3f}")
plt.show()
```

write a function to compute CIs

customize bar chart

14:30 - END
CODING PRACTICALS
PART II